Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 14)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 1 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

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0 1 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]

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0 1 . 3 Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 1 . 4 Explain why representative data are essential for climate modelling.

[2 marks]

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0 2 Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO_2 emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

0 2 . 1 Explain how nuclear fission releases energy in a reactor.

[3 marks]

0 2 . 2 Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

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0 2 . 3 Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

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0 2 . 4 Explain why the CO_2 emissions of nuclear power are not zero.

[1 mark]

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0 2 . 5 Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

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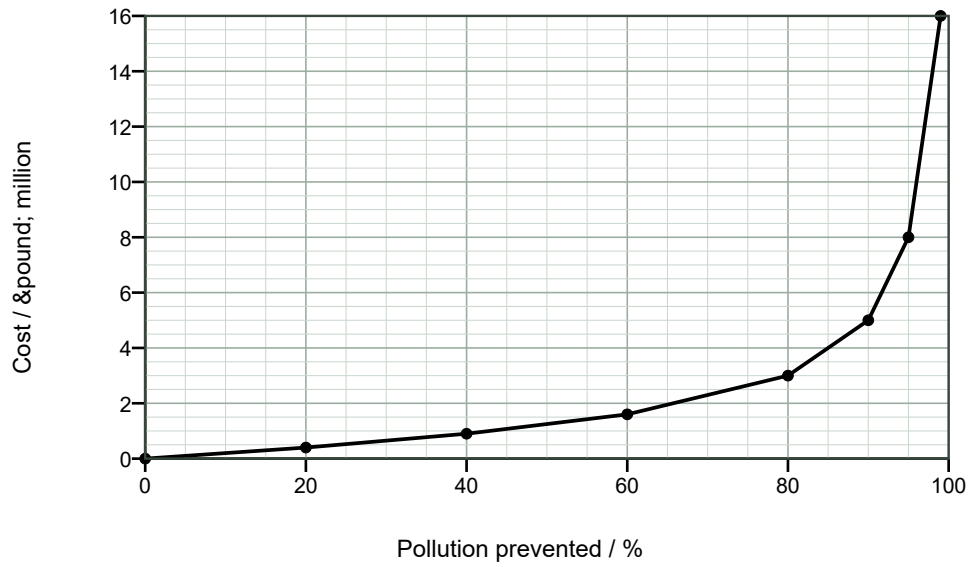
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0 3 Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1



0 3 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 3 . 2 Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]

Answer ☐ £ million

0 3 . 3 Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

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0 3 . 4 Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

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0 4 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 4 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 4 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 5 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 5 . 1 Describe how they could carry out the investigation in the laboratory.

[5 marks]

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0 5 . 2 Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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0 5 . 3 Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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0 6 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 6 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 6 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 7 Table 1 shows the annual mean concentrations of four pollutants measured at a city-centre monitoring station, with the legal limit for each.

Table 1

Pollutant	Annual mean / µg m^{-3}	Legal limit / µg m^{-3}
Nitrogen dioxide	52	40
PM10 particulates	28	40
PM2.5 particulates	19	20
Sulfur dioxide	8	125

0 7 . 1 Calculate by what percentage the annual mean nitrogen dioxide concentration exceeds the legal limit.

[2 marks]

Answer ..... %

0 7 . 2 Suggest why the sulfur dioxide concentration is now far below its limit while nitrogen dioxide is not.

[3 marks]

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0 7 . 3 Explain how each of the following reduces emissions.

[3 marks]

Electrostatic precipitator .....

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Catalytic converter .....

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Low-NOx burner .....

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0 7 . 4 Explain why a reduction in the city's NOx emissions would also reduce summer ground-level ozone.

[3 marks]

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0 7 . 5 The council also plans to monitor air quality using lichens. Explain the advantages and the limitations of using lichens as indicator species.

[4 marks]

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0 8 . 1 Explain how each of the following building design features reduces energy use.

[6 marks]

Orientation .....

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Low surface area : volume ratio .....

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Earth sheltering .....

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0 8 . 2 Compare the embodied energy of concrete with alternative building materials.

[2 marks]

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0 8 . 3 Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

[2 marks]

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0 9 . 1 Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

0 9 . 2 Describe the process of phytomining.

[3 marks]

0 9 . 3 Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

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1 0 . 1 Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

1 0 . 2 Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

11 . 2 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS